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ABSTRACT Vehicles and robots fuse several sensors so that scene understanding survives when one sensor fails, yet the most accurate multimodal segmentation models lose much of their accuracy when a single input is missing. We investigate why. On a synthetic driving benchmark with camera, depth, event, and laser range inputs, we compare a leading fusion model that uses depth supervision with our own model that does not, under one evaluation protocol. Both depend heavily on depth: removing it costs 23 to 39 points of mean intersection over union, more at night than in daylight. When either model is also trained on synthetically degraded inputs, this dependence largely disappears, which indicates that it is learned from clean training data rather than caused by the fusion design. Building on this finding, we present a fusion model on a frozen vision foundation backbone with sensor-specific adapters, symmetric cross-sensor attention, and a high-resolution decoding path, trained with a clean pass and a degraded pass distilled from a clean teacher. The training needs no condition labels and adds no inference cost. The model reaches 57.7 mean intersection over union on the test set, one point above the best published fusion model, and 57.9 under the standard missing-sensor protocol on the validation set, eight points above the best published result. We further analyse sensor dependence across conditions, thin objects, and failure cases [TODO: add one sentence on the real-sensor benchmarks once measured].

INDEX TERMS Foundation models, missing modality, multimodal semantic segmentation, robustness, sensor fusion

I. INTRODUCTION

AUTONOMOUS vehicles and mobile robots carry several sensors because each one fails in different conditions. A camera loses contrast at night and in fog, a depth estimate develops holes on reflective surfaces, an event camera goes silent on static scenes, and LiDAR returns thin out in rain and with distance. Multimodal semantic segmentation promises to cover these failures by fusing the sensors, so that the scene is still understood when one of them degrades.

Research on this problem has split into two tracks. The first track maximizes accuracy when all sensors are present [1]–[6]. The second track measures and improves robustness when sensors are missing or noisy [7]–[11]. The two tracks rarely evaluate the same models under the same protocol. As a result, it is unclear whether the most accurate fusion models are also robust, and if they are not, what makes them fragile. A recent robustness benchmark [10] reports that several fusion models lose much of their accuracy when a sensor is removed, and it explains the differences by fusion design, contrasting RGB-centric, equal-contribution, and adaptive-selection architectures. Whether the design is in fact the cause has not been tested by holding the design fixed and changing only the training.

We study this question on three benchmarks with complementary sensor suites: DELIVER [1] (RGB, depth, event, LiDAR), MUSES [12] (RGB, event, LiDAR, radar), and MCubeS [13] (RGB, polarization, near-infrared). We evaluate a leading accuracy-oriented model, DGFusion [6], and our own fusion model under one protocol, and find that both rely heavily on depth on DELIVER: removing depth costs 23 to 39 points of mIoU, more in the dark than in daylight. DGFusion uses depth supervision, which might explain its dependence, but our model has no such supervision and shows the same pattern. When either model is trained with synthetically degraded inputs in addition to clean ones, the dependence largely disappears. The over-reliance is therefore learned from clean training data rather than built into a specific architecture.

Building on this finding, we present ReliaDINO, a fusion model on a frozen DINOv3 backbone with sensor-specific adapters, symmetric cross-modal attention, and a high-resolution decoding path, trained with a clean pass and a degraded pass that is distilled from a frozen clean teacher. The training scheme needs no condition or degradation labels and

adds no inference cost. On DELIVER, ReliaDINO reaches 57.74 mIoU on the test set (mean of three seeds), 1.0 point above DGFusion, while reducing the depth over-reliance to a third or less, and it scores 57.85 under the standard missing-modality protocol on the validation set, 7.9 points above the best published result [TODO: add the MUSES and MCubeS results once measured].

Our contributions are as follows.

- We show, with controlled comparisons on the same backbone and with re-trained baselines evaluated under one protocol, that the sensor over-reliance of accurate fusion models on DELIVER comes from clean training rather than from their architecture or supervision.
- We combine a frozen foundation backbone with a two-pass training scheme and a high-resolution path into a single model that improves clean accuracy and robustness together, instead of trading one for the other.
- We analyse per-sensor dependence across conditions, missing, partial, and noisy sensors, thin objects, and failure cases, and we report where the model still fails. [TODO: state the code and checkpoint release]

II. RELATED WORK

A. MULTIMODAL FUSION FOR SEGMENTATION

Early RGB-X segmentation fused one auxiliary sensor into an RGB stream, for example by cross-modal feature rectification [2] or by exchanging tokens between modality-specific transformers [14]. CMNeXt [1] generalized this to an arbitrary number of sensors with a hub that aggregates the auxiliary sensors before injecting them into RGB, and introduced DELIVER. Later work improved efficiency and accuracy with adapter-based early fusion [3] and pixel-level token exchange [4]. CAFuser [5] conditions the fusion on the scene condition inferred from the camera image, and DGFusion [6] uses depth, with depth supervision, to guide the fusion; both report leading results on DELIVER and MUSES. These methods are designed and evaluated with all sensors present. We re-train two of them with degraded inputs and show that their robustness depends mostly on training, not on their fusion design.

B. ROBUSTNESS TO MISSING AND DEGRADED SENSORS

A second line of work trains models that tolerate missing sensors. MAGIC [7], [15] treats all modalities equally and, during training, ranks modality features by their similarity to an aggregated feature. AnySeg [8] names the problem unimodal bias, the over-reliance on a dominant, faster-to-learn modality, and reduces it by distilling a multimodal teacher into a student trained with random modality dropout; functional-entropy regularization [16] targets the same bias. RMMSS [9] combines prototype distillation with feature selection. A recent benchmark [10] standardized entire-missing, random-missing, and noisy-modality evaluations, which we adopt, and attributed robustness differences to fusion design. Our study differs from these works in its question and its controls: rather than proposing another robust architecture, we take the accuracy-leading fusion models, hold their architecture fixed, change only the training data, and measure how their dependence on each sensor changes across conditions. FuseAdapt [11] adapts a frozen backbone with parameter-efficient fusion modules for missing modalities. Our training scheme uses the same building blocks as several of these methods, namely synthetic degradation and distillation from a clean teacher. We do not claim it as a new method; we use it to show where over-reliance comes from and to obtain a model that is both accurate and robust. Methods in this line usually skip a missing input, whereas the benchmark protocol replaces it with zeros; we keep the two families apart when comparing numbers.

C. ADAPTING FROZEN FOUNDATION MODELS

Vision foundation models are increasingly used as frozen encoders for dense prediction. The Segment Anything Model [17] has been adapted to multimodal inputs with adapters that inject auxiliary-sensor features into the RGB stream [18] or with mixtures of LoRA experts [19], and MemorySAM [20] reuses the memory mechanism of SAM 2 to fuse modalities. DINOv3 [21] provides dense features and a segmentation adapter derived from ViT-Adapter [22], which we reuse for our high-resolution path. Our backbone, LoRA adaptation [23], and query head [24] follow this practice; we treat them as the platform for our study, not as contributions.

III. SENSORS, BENCHMARKS, AND PROTOCOL

A. SENSORS AND THEIR FAILURE MODES

A driving platform rarely loses all sensors at once; it loses one, partially or completely, under conditions that the other sensors tolerate. An RGB camera gives dense appearance and texture but degrades at night, under low or excessive exposure, and in fog. A depth sensor, or depth reconstructed from stereo or LiDAR, gives geometry that is unaffected by illumination but contains holes on reflective, transparent, and distant surfaces. An event camera reports per-pixel brightness changes with high temporal resolution and dynamic range, but it is silent on static scenes and noisy in low light. A LiDAR gives accurate range on a sparse set of beams that shrinks with distance and is thinned by rain, fog, and absorbing surfaces. Radar, available in one of our benchmarks, measures range and velocity through adverse weather at a very low angular resolution. Each failure is therefore local to one sensor and correlated with a scene condition, which is the situation that a fusion model has to handle without being told which sensor has failed.

B. BENCHMARKS

We evaluate on three public benchmarks that cover synthetic and real sensor suites (Table 1). DELIVER [1] is a synthetic driving benchmark with RGB, depth, event, and LiDAR inputs and 25 classes. Its scenes span five weather and lighting conditions (sun, cloud, rain, fog, and night) and five sensor-failure cases (motion blur, over- and under-exposure, LiDAR jitter, and low-resolution events). Because every scene is

TABLE 1. Benchmarks used in this study.

Benchmark	Sensors	Classes	Train / val / test
DELIVER [1]	RGB, depth, event, LiDAR	25	3983 / 2005 / 1897
MUSES [12]	RGB, event, LiDAR, radar	19	1500 / 250 / 750
MCubeS [13]	RGB, AoLP, DoLP, NIR	20	todo / todo / 102

MUSES test labels are withheld; test scores come from the official evaluation server.

rendered, the depth input is complete and noise-free, which turns out to matter for how fusion models learn (Section VI). MUSES [12] is a real driving dataset recorded under clear, foggy, rainy, and snowy conditions by day and night, with an RGB frame camera, an event camera, a MEMS LiDAR, and an FMCW radar, all projected into the camera view, and the 19 Cityscapes classes. We use all four sensors. MCubeS [13] is a real material-segmentation dataset with RGB, angle and degree of linear polarization (AoLP, DoLP), and near-infrared (NIR) images and 20 material classes. It tests whether the same model transfers to non-driving sensors whose failure modes differ from those above.

C. EVALUATION PROTOCOL

Clean accuracy. We report mean intersection over union (mIoU). For each training run we keep the checkpoint with the best validation score and report its test score, without selecting checkpoints on the test set. On DELIVER we evaluate at 1024×1024 with batch size one against the native-resolution labels, and resample labels with nearest-exact interpolation. This detail matters: floor-aligned nearest resampling shifts predictions by half a pixel and lowered DELIVER test mIoU by 1.2 to 1.3 points in our measurements, with larger losses on thin classes, whereas center-aligned resampling reproduces the scores of the baselines' own conversion code. We therefore use center-aligned resampling for every model.

Missing and degraded sensors. We follow the definitions of a recent robustness benchmark for multimodal segmentation [10]. In the entire-missing-modality (EMM) setting, one or more sensors are replaced by zeros after input normalization, and the score is averaged over all 15 non-empty subsets of the four sensors. In the random-missing-modality (RMM) setting, the sensors that a subset would remove are instead corrupted by zeroing a random fraction $r \in \{0.25, 0.5, 0.75\}$ of their values, and the score is again averaged over the 15 subsets. In the noisy-modality (NM) setting, salt-and-pepper noise of density $D \in \{0.05, 0.1, 0.2\}$ corrupts every sensor at the same pixel positions. The benchmark additionally adds Gaussian noise in this setting; we report salt-and-pepper alone so that the corruption is identical for all sensors [TODO: confirm that every NM number in Table 5 was produced with the salt-and-pepper-only mode (tools/missing_modality_eval.py, default nm mode)].

IV. METHOD

Fig. 1 summarizes ReliaDINO. The model keeps a single foundation backbone frozen and shares it across all sensors, so that every sensor is read with the same visual vocabulary. It adds a small number of trained components around the backbone: sensor-specific low-rank adapters, a symmetric cross-modal fusion, a decoder with a high-resolution path, and two complementary heads. Training uses every batch twice, once clean and once with synthetic sensor degradation, and distills a frozen clean-trained teacher into the degraded pass.

A. FROZEN BACKBONE WITH SENSOR-SPECIFIC ADAPTERS

Let x_s denote the input of sensor $s \in \{1, \ldots, S\}$, converted to a three-channel image (Section V). Every x_s is processed by the same DINOv3 ViT-L/16 backbone [21], whose weights are frozen. To let each sensor adapt the shared features to its own statistics, we attach a separate low-rank adapter (LoRA) [23] of rank 8 to the attention projections of the backbone for each sensor. Instead of reading only the last layer, we take the token maps after blocks 6, 12, 18, and 24, so that the decoder receives both low-level and semantic evidence; each of the four maps is projected by a sensor-specific 1×1 layer. The result is a stride-16 token set T_s for every sensor.

B. SYMMETRIC CROSS-MODAL FUSION

Many fusion designs nominate RGB as the primary stream and inject the other sensors into it [2], [18], which makes the model fragile exactly when RGB fails. We instead treat the sensors symmetrically. In each of two fusion layers, the tokens of sensor s form the queries of a multi-head attention whose keys and values are the tokens of all other sensors,

$$T_s \leftarrow T_s + \mathrm{MHA}\big(T_s,\ [T_{s'}]_{s' \neq s},\ [T_{s'}]_{s' \neq s}\big), \qquad (1)$$

followed by a feed-forward layer. No sensor is privileged, and the attention has no gate and receives no external reliability signal. The fused tokens of all sensors are merged by a gated MLP into one stride-16 feature map F. In addition, a per-class sensor router inherited from earlier versions of our model adds a zero-initialized residual term to the logits. [TODO: decide with the method session how to describe the router (MODEL.ROUTER = per-class reliability-anchored modality router, fusion.py:64) and whether it needs its own ablation row; earlier internal measurement: about +0.1 mIoU]

C. DECODER WITH A HIGH-RESOLUTION PATH

A SimpleFPN [22] turns F into a feature pyramid at strides 4, 8, 16, and 32. Because every level is interpolated from stride-16 tokens, structures thinner than a token, such as poles and pedestrians at a distance, are poorly represented. We therefore add a high-resolution path that follows the adapter used in the official DINOv3 segmentation code [21], [22]. A convolutional stem with weights shared across sensors (and a sensor-specific 1×1 projection) extracts multi-scale features from the raw inputs. Four deformable-attention extractors then let these convolutional queries sample semantic evidence

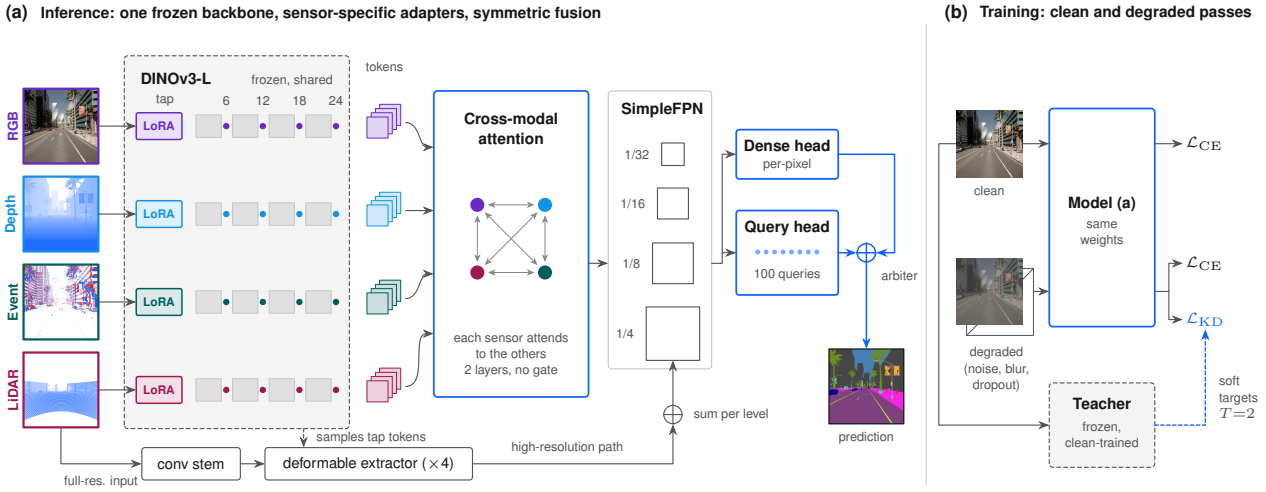


FIGURE 1. Overview of ReliaDINO. (a) Each sensor passes through the same frozen DINOv3-L with its own LoRA adapter; tokens from four depths are fused by symmetric cross-modal attention, decoded by a SimpleFPN augmented with a high-resolution path, and classified by a dense head and a query head combined by an arbiter. (b) Every batch is used twice: a clean pass and a synthetically degraded pass, the latter also distilled from a frozen teacher trained on clean data. Blue outlines mark trained components; dashed boxes are frozen. The scene and the prediction are a DELIVER test image and the output of our model. [TODO: draft drawn by the paper session; redraw per notes/figure_guide.md]

from the four tapped token maps. Unlike the original adapter, we use only the extractors and do not inject anything back into the backbone, which stays untouched. The path's output is added to the FPN levels without a gate. We regard this path as an engineering complement taken from prior work, not as a contribution, and we analyze where its gain comes from in Section VI.

D. DENSE AND QUERY HEADS

Two heads read the pyramid. A dense convolutional head classifies every pixel at stride 4. A Mask2Former-style query head [24] with 100 queries and six decoder layers predicts class scores and masks, which are converted into per-pixel class scores Q. An arbiter combines the two with a learned per-class weight λ_k that is initialized at zero,

$$\ell_k(p) = D_k(p) + \lambda_k \, Q_k(p), \tag{2}$$

so that the query head contributes only where it helps the dense prediction.

E. TWO-PASS TRAINING WITH TEACHER DISTILLATION

Our analysis in Section VI shows that over-reliance on one sensor is learned from clean training data. We therefore change the training signal rather than the architecture. Every batch (x, y) is used twice. The clean pass computes the segmentation loss $\mathcal{L}_{\text{seg}}(f(x), y)$. The degraded pass builds $\tilde{x}$ by corrupting each sensor of each sample independently with probability 0.5; one fifth of the corrupted sensors are removed entirely (zero-filled), and the rest receive one operator drawn from a sensor-specific set: patch dropout, blur, gain and gamma changes, colour shift (RGB), depth holes (depth), beam dropout (LiDAR), and resolution loss (event). The corruption strength follows a curriculum that rises from 0.3 to 0.6 and 1.0 of its maximum at epochs 0, 10, and 20. Gaussian and salt-and-pepper noise are deliberately excluded from training so that the noise evaluation tests generalization to unseen corruption.

The degraded pass is supervised by the labels and by a frozen teacher g, a model of the same architecture without the high-resolution path that was trained on clean data only. The teacher sees the clean input, and its soft predictions are distilled into the degraded pass:

$$\begin{aligned}\mathcal{L} = \mathcal{L}_{\text{seg}}(f(x), y) + \mathcal{L}_{\text{seg}}(f(\tilde{x}), y) \\ + \alpha \, T^2 \, \mathrm{KL}\left(\sigma\left(\frac{g(x)}{T}\right) \middle\| \sigma\left(\frac{f(\tilde{x})}{T}\right)\right),\end{aligned} \tag{3}$$

with $\alpha = 0.5$ and $T = 2$, where σ is the softmax over classes and the Kullback–Leibler divergence is averaged over pixels. The scheme needs no condition labels and no degradation labels, and it adds no computation at inference. Its parts are known: clean-teacher distillation and synthetic dropout appear in robust multimodal segmentation [8], [9]. Our point is not a new loss but the evidence that this change of training signal, and not the architecture, removes the sensor over-reliance (Section VI).

V. EXPERIMENTS

A. IMPLEMENTATION DETAILS

All sensors are converted to three-channel images as in CMNeXt [1]. We train for 40 epochs with AdamW (learning rate 6×10^{-4}, weight decay 0.01), a polynomial schedule with power 0.9, and a 10-epoch warm-up, on 768×768 crops with an effective batch size of 16 obtained by gradient accumulation. We use the data augmentation of DGFusion [6] so that both models see the same augmented images. The segmentation loss combines online hard-example cross-entropy [25] on the dense head with the set-prediction loss of the query head (weight 0.5, with deep supervision) and

TABLE 2. DELIVER (RGB, depth, event, LiDAR), mIoU (%).

Method	Backbone	Sensors	val	test
CMNeXt todo	todo	RGB-D-E-L	todo	todo
StitchFusion todo	todo	RGB-D-E-L	todo	todo
GeminiFusion todo	todo	RGB-D-E-L	todo	todo
CAFuser todo	todo	RGB-D-E-L	todo	todo
CAFuser-CAA todo	todo	RGB-D-E-L	68.79	todo
DGFusion todo	Swin-T	RGB-D-E-L	todo	56.71
Fewer sensors (secondary comparison)				
MM SAM-adapter todo	todo	RGB-D	69.60	57.35
Ours (3-seed mean)	DINOv3-L	RGB-D-E-L	todo	57.74
Ours (best seed)	DINOv3-L	RGB-D-E-L	todo	58.61

todo: original table/page per baseline; protocol (val-best checkpoint, 1024, nearest-exact label resampling); DINOv3-L is frozen.

TABLE 3. MUSES, official evaluation, mIoU (%).

Method	Backbone	Sensors	val	test
CAFuser todo	todo	todo	todo	todo
DGFusion todo	todo	todo	todo	79.5
Camera only or fewer sensors (secondary comparison)				
GtA todo	todo	RGB	todo	82.39
MM SAM-adapter todo	todo	RGB-L	todo	81.07
Ours	DINOv3-L	RGB-E-L-R	todo	todo

TABLE 4. MCubeS (RGB, AoLP, DoLP, NIR), test mIoU (%).

Method	Backbone	test
Mul-VMamba todo	todo	54.65
StitchFusion todo	todo	55.9
Ours	DINOv3-L	todo

TABLE 5. Robustness to missing and degraded sensors on DELIVER, mIoU (%).

(a) Public missing-modality benchmark, val

Method	EMM	RMM .25	RMM .50	RMM .75
RMMSS todo	49.93	todo	todo	todo
Ours (3 seeds)	57.85	65.53	63.35	61.28

(b) Same protocol for all models, test

Method	Clean	EMM	S&P .05	S&P .10	S&P .20
DGFusion (official)	55.68	31.53	37.97	16.71	5.52
DGFusion (retrained†)	54.77	45.52	48.89	40.51	23.34
CAFuser (retrained†)	55.79	46.50	todo	todo	todo
Ours w/o two-pass	todo	40.01	todo	todo	todo
Ours w/o high-res	56.59	48.06	53.25	51.64	48.79
Ours	57.74	48.66	53.11	51.36	47.85

†Retrained with the same synthetic degradation. Ours: mean of 3 seeds; baselines: single runs. todo: define EMM (mean over 15 sensor subsets), RMM (random drop ratio), S&P noise.

three auxiliary terms inherited from earlier versions of our model: an auxiliary cross-entropy on the fusion output, a class-prototype contrastive term (weight 0.1), and a variance-covariance regularizer on the fused tokens; we list them for reproducibility and do not claim them as contributions [TODO: one-line ablation or citation for the auxiliary terms, or move to the supplement]. The frozen teacher is the same architecture without the high-resolution path, trained for 200 epochs on clean data. Unless noted, we report the mean of three seeds and the best seed; ablations use one seed.

B. COMPARISON WITH PRIOR FUSION MODELS

Table 2 compares ReliaDINO with fusion models on DELIVER. We list only methods evaluated under the protocol that reports test-split mIoU, because DELIVER results in the literature are reported under at least three different protocols and are not comparable across them. ReliaDINO reaches 57.74 test mIoU on average over three seeds and 58.61 for the best seed, 1.0 and 1.9 points above DGFusion. Its margin over MM SAM-adapter, which uses only RGB and depth, is 0.4 points on average [TODO: val 3-seed mean; val comparison with CAFuser-CAA 68.79]. Tables 3 and 4 report the same model on the real MUSES and MCubeS benchmarks [TODO: results pending: MUSES four-sensor model, MCubeS transfer].

C. ROBUSTNESS TO MISSING AND DEGRADED SENSORS

Table 5 and Fig. 2 evaluate all models under the same missing-sensor and noise protocol. Under the public benchmark definition on the validation set, ReliaDINO scores 57.85 EMM, 7.9 points above the best published result, and it keeps 61 to 66 mIoU when a quarter to three quarters of a sensor's pixels are missing. On the test set, the official DGFusion weights fall to 31.5 EMM and to 5.5 mIoU under the strongest noise, which shows how much the accuracy-oriented models depend on complete inputs. Re-training DGFusion and CAFuser with our synthetic degradation raises their EMM to 45.5 and 46.5 without changing their architecture. ReliaDINO reaches 48.7 EMM while keeping the highest clean accuracy, so it does not trade accuracy for robustness, and it retains 47.9 mIoU under the strongest noise, which was never seen during training.

When only one sensor is available (Fig. 2b), the official DGFusion collapses to 12.1 mIoU with RGB alone and to 1.2 and 3.2 with event or LiDAR alone, while still reaching 43.1 with depth alone. Training with degraded inputs lifts every single-sensor score, most strongly for RGB (49.0 for the retrained DGFusion and 53.9 for our model without the high-resolution path). Sensor dependence, in other words, follows the training data.

D. QUALITATIVE RESULTS

Figs. 3 and 4 show scenes selected by a fixed rule: among all test images, those where the gap between our model and the best baseline is largest, at most one scene per condition and map [TODO: describe what the scenes show once selected].

E. ACCURACY PER CONDITION

Table 6 breaks clean test accuracy down by condition [TODO: text after the per-condition numbers arrive (analysis request

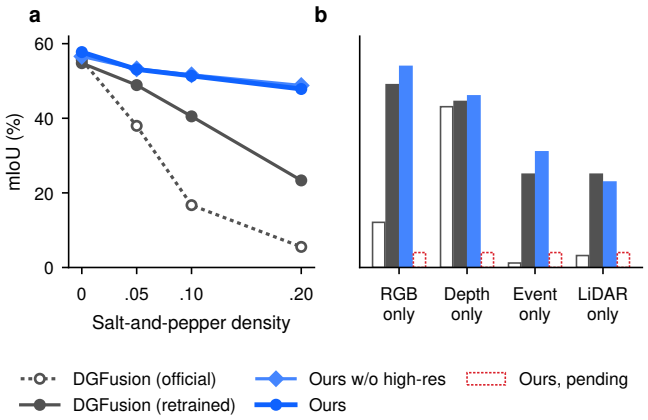


FIGURE 2. **Robustness on DELIVER test under one protocol. (a) Salt-and-pepper noise applied to every sensor. (b) Only one sensor available (the other three zero-filled). Ours: mean of three seeds; DGFusion: single runs, official weights or retrained with the same synthetic degradation. [TODO: (b) full-model values pending.]**

TABLE 6. **DELIVER test mIoU (%) per condition, all sensors present.**

Model	Sun	Cloud	Rain	Fog	Night	Low exp.	All
DGFusion (official)	todo	todo	todo	todo	todo	todo	55.68
DGFusion (retrained†)	todo	todo	todo	todo	todo	todo	54.77
CAFuser (retrained†)	todo	todo	todo	todo	todo	todo	55.79
Ours w/o high-res	todo	todo	todo	todo	todo	todo	56.59
Ours	todo	todo	todo	todo	todo	todo	57.74

†Retrained with synthetic degradation. All: same values as Table 5(b) clean column.

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F. ABLATION

Table 7 builds the model up from a frozen DINOv3-L with LoRA on RGB alone and then removes one component at a time from the full model, all with the same seed and schedule. The frozen backbone alone reaches 55.14 mIoU with RGB, already within 1.6 points of DGFusion with four sensors, so a large part of the accuracy comes from the foundation model. Adding the other three sensors with cross-modal attention and the query head gives 55.94, the two-pass training with distillation adds 1.8 points (57.73), and the high-resolution path adds 0.9 (58.61). Removing cross-modal attention or the query head costs 0.85 and 0.72 points, respectively. Replacing the sensor-specific adapters with one shared adapter does not reduce clean accuracy (58.84) [TODO: robustness of the shared variant and the degradation-pass and distillation rows (L5, L6) pending].

G. COMPUTATIONAL COST

Table 8 reports parameters, operations, latency, and memory at the evaluation resolution [TODO: numbers pending (DRN-261027-02); the two-pass training doubles the forward passes during training but adds nothing at inference].

VI. ANALYSIS

TABLE 7. **Ablation on DELIVER test (one seed, 40 epochs), mIoU (%).**

Configuration	Clean	EMM
Building up		
Frozen DINOv3-L + LoRA, RGB only	55.14	–
+ depth, event, LiDAR (mean fusion)	todo	todo
+ cross-modal attention, query head	55.94	todo
+ two-pass training, distillation	57.73	48.68
+ high-res path (full model)	58.61	49.12
Removing one component from the full model		
− sensor-specific LoRA (one shared)	58.84	todo
− cross-modal attention (mean fusion)*	57.76	todo
− query head and arbiter	57.89	todo
− degradation pass	todo	todo
− teacher distillation	todo	todo

*Also removes the trunk expansion. todo: EMM column for ablation runs; decision on shared LoRA pending its robustness test.

TABLE 8. **Computational cost on DELIVER at 1024 × 1024.**

Model	Params (M)	Trainable (M)	GFLOPs	Latency (ms)	Memory (GB)
DGFusion	todo	todo	todo	todo	todo
CAFuser	todo	todo	todo	todo	todo
Ours w/o high-res	todo	todo	todo	todo	todo
Ours	todo	todo	todo	todo	todo

A. WHERE SENSOR OVER-RELIANCE COMES FROM

A fusion model can lean on one sensor because its architecture or supervision favours that sensor, or because the training data make that sensor sufficient. DGFusion [6] is a useful test case: it adds depth supervision to its fusion, and removing depth costs the official model 23 to 37 points depending on the condition (Fig. 5). If the depth supervision caused this dependence, a model without it should behave differently.

Our own architecture trained only on clean data, which has no depth supervision, shows the same pattern: removing depth costs 23.5 to 38.7 points, and the cost grows from sun to low exposure exactly as for DGFusion. The same architecture trained with the two-pass scheme loses only 3.9 points in sun, 10.1 at night, and 13.4 under low exposure. The effect also holds across architectures. When we re-train DGFusion with the same synthetic degradation, removing depth entirely costs 4.7 points instead of 33.3, and the EMM score rises from 31.5 to 45.5 (Table 5).

We conclude that, on DELIVER, the dependence on depth is a property of clean training rather than of a particular fusion design or supervision. A plausible reason is that DELIVER renders depth without noise or holes, so clean training finds a shortcut that real depth sensors would not offer. Two caveats bound this conclusion. The clean-only model was trained for 200 epochs and the two-pass model for 40, and the DGFusion curve was measured on the test split while ours were measured on validation [TODO: remove the second caveat once all curves are re-measured on one split]. A complementary observation is that our clean-only model is already less sen-

todo: Fig. 3 when a sensor fails: rows = depth missing / RGB missing / LiDAR+event missing / RGB with heavy noise; columns = affected input, ground truth, DGFusion (official), DGFusion (retrained), CAFuser (retrained), ours; error pixels in one colour

FIGURE 3. **[TODO: caption; state the scene-selection rule (top differences between ours and the best baseline, one scene per condition, no repeated map)]**

todo: Fig. 4 adverse conditions, all sensors present: DELIVER night / fog / rain / low exposure and MUSES real night / snow / fog; columns = RGB, ground truth, DGFusion, CAFuser, ours; zoomed crops where the baselines miss objects

FIGURE 4. **[TODO: caption; state the scene-selection rule]**

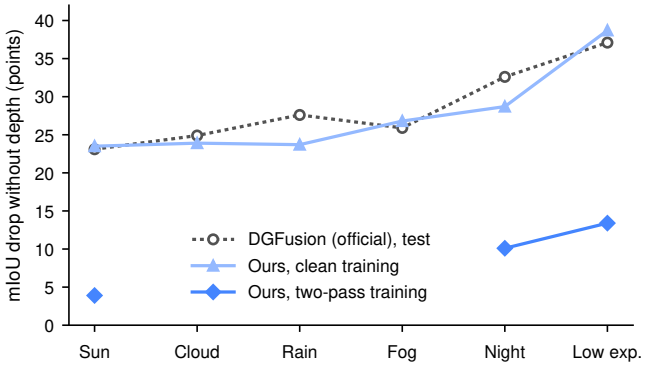


FIGURE 5. **Drop in mIoU when depth is removed, per condition. A model trained only on clean data relies on depth as heavily as DGFusion, whose design adds depth supervision; two-pass training reduces the drop to a third or less with the same architecture. DGFusion was measured on test, the other curves on val. [TODO: re-measure all curves on one split; add the full model and the retrained baselines.]**

sitive to salt-and-pepper noise than the re-trained DGFusion; we therefore attribute part of the noise robustness to the frozen DINOv3 backbone rather than to our training [TODO: state the measured drops (ledger 2026-10-01: E1 3.09/5.73/11.22 vs DGFusion retrained 5.9/14.3/31.4) after checking that the definitions match].

B. WHAT THE HIGH-RESOLUTION PATH ADDS

The high-resolution path raises clean test mIoU by 1.15 points on average over three seeds (from 56.59 to 57.74; Table 7). Fig. 6 shows where the gain comes from. Averaged over the three seeds, Static gains 10.4 points, Pedestrian 3.8, TrafficSign 3.1, and Pole 2.1, and the four thin classes gain 3.0 points together. The gain comes mainly from recall: for Pole the recall rises from 0.75 to 0.78 and for Pedestrian from 0.89 to 0.90 [TODO: confirm these recall values (two seeds, NAS analysis_logs/a2_thin_anatomy_20261018/summary.md) through the analysis session before citing]. Two classes lose: TrafficLight by 4.1 points and RailTrack by 7.2. RailTrack varies strongly between seeds even without the path (58.5, 32.6, and 45.1 for the three seeds), so we do not interpret its change. The TrafficLight loss, in contrast, is systematic and is analysed next.

C. FAILURE CASE: POSTERS READ AS TRAFFIC LIGHTS

With the high-resolution path, the recall of TrafficLight rises (0.80 to 0.85) but its precision falls (0.46 to 0.41) [TODO: same confirmation as above]. Fig. 7 shows where the extra false positives appear. In the scenes with the largest increase, billboards and posters, labelled as *Other*, are painted as traffic lights, and in one rainy scene a traffic sign is. The two classes

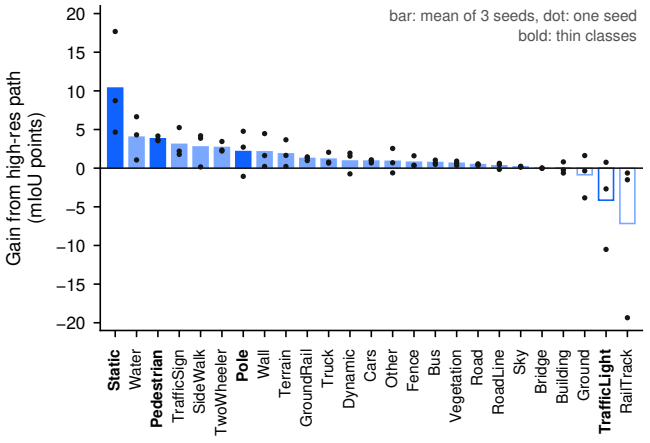


FIGURE 6. **Per-class gain of the high-resolution path on DELIVER test, paired by seed (full model minus the same model without the path). Thin classes gain most; TrafficLight and RailTrack lose and vary strongly across seeds. [TODO: add zoomed crops of thin objects (Q2 dumps on yeon).]**

todo: limitation: posters and signs predicted as traffic light (2 of the 4 recorded scenes; RGB, ground truth, ours, error map)

FIGURE 7. **[TODO: caption; selection rule from the scene README: among the 20 test images where our model adds the most TrafficLight false positives relative to the model without the high-resolution path, one scene per condition.]**

share fine structure, a small bright rectangular shape on a pole or a wall, but differ in context. Because the high-resolution features are added to every pyramid level without a gate, fine detail reaches the dense per-pixel classifier directly, and the classifier can follow the detail without enough context. This mechanism is a hypothesis that our measurements are consistent with but do not prove; we discuss a remedy in Section VII.

D. LIMITATIONS

First, DELIVER is synthetic, and its complete depth is the reason the shortcut exists; the size of the effect on real depth sensors remains to be measured, and MUSES does not contain depth. Second, all our runs use a 40-epoch schedule and three seeds, while the baselines are single runs, so differences below about one point should not be over-read. Third, the high-resolution path makes the model slightly more sensitive to heavy pixel noise and to partial loss of depth or RGB than the same model without it (Table 5); a likely cause is that its convolutional stem sees raw pixels, which we have not yet verified. Fourth, a single shared LoRA reached the same clean accuracy as sensor-specific adapters in one seed (Table 7); [TODO: update once the shared-LoRA robustness test decides which variant is the paper model].

VII. CONCLUSION

We asked why accurate multimodal segmentation models fail when one sensor degrades. Controlled comparisons on DELIVER showed that their heavy dependence on depth is learned from clean training data: it appears in our model without depth supervision as strongly as in DGFusion with it, and it largely disappears in both architectures when degraded inputs are added to training. Built on this finding, ReliaDINO combines a frozen DINOv3 backbone, symmetric fusion, and a high-resolution path with two-pass training, and improves clean accuracy and robustness together on DELIVER [TODO: and on MUSES and MCubeS, once measured]. The analysis also exposes what remains: the high-resolution path recovers thin objects but lets fine detail mislead the per-pixel classifier, as in posters read as traffic lights, which suggests routing fine detail to mask prediction rather than to per-pixel classification. Measuring the same dependence on real depth sensors is the next step that our synthetic-data findings call for.

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[TODO: AUTHOR NAME] [TODO: biography (photo 1 in × 1.25 in)]